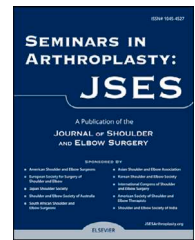


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The influence of insurance type on achievement of patient-reported outcome thresholds in anatomic and reverse shoulder arthroplasty

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ABSTRACT

Background: Patient-reported outcome measures (PROMs) for total shoulder arthroplasty are commonly collected for both clinical and research utilization and have become increasingly of interest to policy and regulatory figures. With this heightened focus, we sought to assess whether payer type affects the achievement of PROM thresholds. We hypothesized that payer type would influence achievement of PROM thresholds, specifically that government insurances would be associated with lower rates of achievement.

Methods: Patients who underwent either anatomic or reverse total shoulder arthroplasty (aTSA and rTSA, respectively) between July 2022 and April 2024 and completed preoperative and 1-year American Shoulder and Elbow Surgeons (ASES) Scores were collected. Self-pay and Medicaid status patients were excluded from analysis due to low counts. Remaining patients were grouped by Commercial, Traditional Medicare, and Medicare Advantage status. The ASES aTSA thresholds were the minimum clinically important difference (MCID), substantial clinical benefit (SCB), and the patient acceptable symptom state (PASS) at 14.9, 35.2, and 82.6, respectively. For rTSA, the MCID, SCB, and PASS thresholds were 13.4, 31.5, and 77.5, respectively. Rates of achievement of thresholds were calculated per procedure and payer. Descriptive analytical statistics and multivariate logistic regressions were conducted.

Results: In total, 81 aTSA and 81 rTSA patients were included. No threshold achievement rate differences were observed in aTSA, but rTSA PASS achievement was statistically higher in Commercial payers (89.47%) compared to Traditional Medicare and Medicare Advantage at 45.95 and 48.00%, respectively; $P = .0044$. Upon multivariate regression, Traditional Medicare was associated with a 79% less likelihood of achieving MCID ($P = .0070$) and a 76% less likelihood to achieve SCB ($P = .0070$). Both Medicare Advantage and Traditional Medicare were statistically associated with decreased likelihood of achieving PASS (73 and 67% decreases; $P = .0226$ and 0.0272 , respectively).

Conclusion: The achievement of clinically relevant PROM thresholds was statistically different among payer types, with Traditional Medicare patients experiencing significant reductions in likelihood of achieving these important recovery milestones when controlling for potentially relevant factors such as age and gender. In the era of value-based care, disparities in the achievement of clinically meaningful PROMs across payer types highlight the need for further research and targeted intervention to ensure equitable recovery and outcome measurements for all patients.

Level of evidence: Level IV; Retrospective Case Series

Keywords: Shoulder arthroplasty; Insurance type; Patient-reported outcome measures; Minimum clinically important difference; Healthcare disparities; Value-based reimbursement

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The Texas Orthopedic Hospital institutional review board approved this study (TOH265e).

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<https://doi.org/10.1016/j.sart.2026.151544>

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Total shoulder arthroplasty, both anatomic (aTSA) and reverse (rTSA), have significantly increased in utilization in the past decade, largely due to the aging population of the United States.^{9,19} Indications for shoulder arthroplasty include glenohumeral arthritis, rotator cuff arthropathy, irreparable rotator cuff tears, inflammatory arthropathy, post-traumatic deformities, and osteonecrosis. rTSA is specifically indicated over total shoulder arthroplasty when the rotator cuff is dysfunctional or there is significant bone loss.⁶ Patients who undergo rTSA are generally older, as they are more likely to have rotator cuff insufficiency or advanced bone loss.⁵ For patients with rotator cuff deficiency or significant boney deformity, the rTSA provides a more biomechanically favorable architecture to the joint and optimizes deltoid pull.^{4,13}

As literature continues to evolve regarding long-term postoperative outcomes, it is increasingly important to investigate factors that may influence these outcomes, including payer type. Most shoulder arthroplasties are covered by insurance, with higher utilization of Medicare (66%) compared to Commercial insurance (23%).⁸ Payer type often serves as a proxy for socioeconomic status (SES), as it is closely tied to employment, financial stability, and access to health care. Lower SES has been associated with poorer outcomes in lower-extremity joint arthroplasty,¹² with similar findings in shoulder arthroplasty.² Age is also associated with Medicare status, as eligibility is determined by patient age.

This study evaluates the potential relationship between payer type and patient-reported outcome measures (PROMs). Assessing PROMs using thresholds of achievement such as the minimum clinically important difference (MCID), substantial clinical benefit (SCB), and patient acceptable symptom state (PASS) provides a multidimensional understanding of perceived functional outcomes from the patient's perspective. By comparing these outcome measures across Medicare subgroups and commercially insured, this study aims to identify any disparities in postoperative recovery associated with payer type. Understanding such differences is critical for identifying at-risk patient populations and informing efforts to improve equity in surgical outcomes. We hypothesize that payer type influences the likelihood of achieving clinically meaningful PROM thresholds, specifically that government payers would achieve lower thresholds.

Methods

This institutional review board–approved (TOH265e), retrospective study utilized an internal database to collect all aTSA and rTSA procedures that occurred between July 2022 and April 2024 at a single site by 1 of 5 orthopedic surgeons. For inclusion, patients were required to have completed all preoperative and 1-year postoperative PROMs, which included the ASES, Single Assessment Numeric Evaluation (SANE), and Patient-Reported Outcomes Measurement Information System Global-10 (PROMIS-10) surveys. Patient information such as age, gender, and payer was collected from electronic medical records. Payers were classified into 3 cohorts:

Commercial, Traditional Medicare, and Medicare Advantage. All other payer types, such as self-pay, workers' compensation, and Medicaid, were excluded due to extremely low rates of occurrence inconducive to analysis (See Fig. 1).

The ASES MCID, SCB, and PASS thresholds were sourced from literature for both aTSA and rTSA.¹⁶ The aTSA MCID, SCB, and PASS thresholds were 14.9, 35.2, and 82.6, respectively; and the rTSA MCID, SCB, and PASS thresholds were 13.4, 31.5, and 77.5, respectively.¹⁶ The Δ ASES (difference between 1 year postoperative and preoperative score) was calculated and compared to the MCID and SCB thresholds. The 1-year postoperative ASES score was compared to the PASS score. The percentage of patients achieving the 3 PROM thresholds were calculated by procedure and payer and statistically analyzed. Categorical variables were assessed with a χ^2 test, and continuous variables with a t-test. Multivariate logistic regressions were performed on the collective cohort and assessed the following variables for influence on achieving MCID, SCB, and PASS: age, gender, aTSA, rTSA, and payer. Odds ratios and confidence intervals of 95% were calculated for the results of the regressions. The results of all tests were determined as significant if $P < .05$.

Results

Descriptive overview

In total, 81 aTSAs and 81 rTSAs were included in the study. The patient selection process and groupings by payer can be found in Figure 1. Age was statistically significant different among the payers in both aTSA and rTSA, where commercial patients were youngest ($P < .0001$; Table 1). There were no statistically significant differences of patient gender representation in either procedure with respect to the payers. Additionally, no differences in any preoperative PROMs were observed in either procedure. The rates of achieving MCID, SCB, or PASS were not significant for aTSA irrespective of payer. Commercial, Traditional Medicare, and Medicare Advantage achieved MCID at 90.63, 84.49, 91.67%; $P = .8156$, respectively, SCB at 68.75, 54.05, and 91.67; $P = .0517$, respectively, and PASS at 62.50, 64.86, and 50.00%; $P = .6502$, respectively. For rTSA, no significant differences among achievements by payer were observed except that PASS achievement was statistically increased in Commercial payers (89.47%) compared to Traditional Medicare and Medicare Advantage at 45.95 and 48%, respectively; $P = .0044$. MCID achievement by Commercial, Traditional Medicare, and Medicare Advantage was 100, 75.68, and 92.00%; $P = .0762$, and SCB achievement was 89.47, 59.46, and 64.00; $P = .0658$, respectively.

Logistic regressions

Traditional Medicare was significantly associated with a 79% less likelihood of achieving MCID ($P = .0070$), and no other parameters were found significant after controlling for other

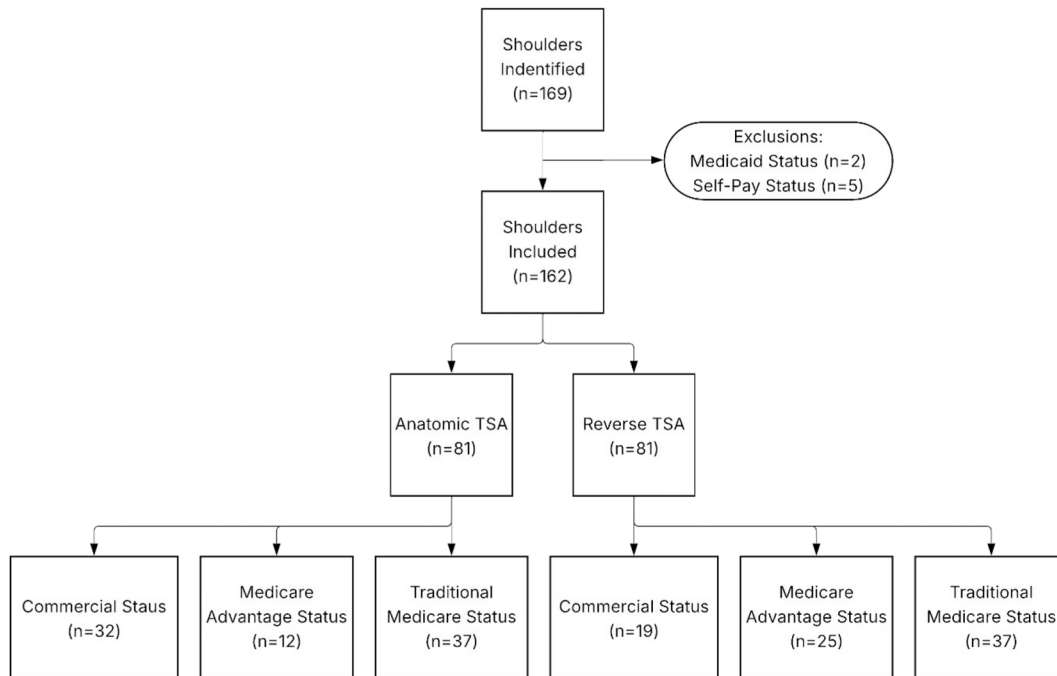


Figure 1 – Patient selection flow diagram for inclusion by procedure type and payer type. TSA, total shoulder arthroplasty.

Table I – Demographics and general comparisons among payer categories by procedure.

Anatomical TSA				
Parameter	Commercial (n = 32)	Traditional Medicare (n = 37)	Medicare advantage (n = 12)	P value
Age (Yr)	60.66 ± 5.10	73.03 ± 6.63	73.67 ± 7.60	<.0001
Female (%)	12 (37.50)	22 (59.46)	8 (66.67)	.1027
Preoperative ASES	38.94 ± 15.40	41.14 ± 15.68	31.92 ± 10.66	.1855
Preoperative SANE	31.97 ± 21.17	33.32 ± 17.11	37.33 ± 29.54	.7505
Preoperative PCS	43.77 ± 6.41	43.69 ± 6.07	43.17 ± 5.46	.9568
Preoperative MCS	52.94 ± 8.33	52.13 ± 7.17	53.36 ± 7.23	.8527
1 yr ASES	83.69 ± 16.00	81.22 ± 21.70	82.00 ± 19.38	.8673
1 yr SANE	80.19 ± 19.34	76.24 ± 26.24	80.75 ± 20.72	.7251
1 yr PCS	49.14 ± 8.16	47.94 ± 8.45	48.77 ± 8.94	.8362
1 yr MCS	53.31 ± 8.43	51.86 ± 7.63	52.76 ± 8.50	.7564
Met MCID (%)	29 (90.63)	32 (86.49)	11 (91.67)	.8156
Met SCB (%)	22 (68.75)	20 (54.05)	11 (91.67)	.0517
Met PASS (%)	20 (62.50)	24 (64.86)	6 (50.00)	.6502
Reverse TSA				
Parameter	Commercial (n = 19)	Traditional Medicare (n = 37)	Medicare advantage (n = 25)	P value
Age (Yr)	63.05 ± 8.70	71.19 ± 6.44	72.96 ± 5.53	<.0001
Female (%)	8 (42.11)	18 (48.65)	16 (64.00)	.3084
Preoperative ASES	37.47 ± 17.31	37.11 ± 20.30	31.24 ± 18.69	.4322
Preoperative SANE	37.42 ± 20.35	33.65 ± 22.87	28.60 ± 24.54	.4382
Preoperative PCS	44.08 ± 5.99	42.69 ± 7.76	41.02 ± 9.10	.4329
Preoperative MCS	53.11 ± 8.71	50.18 ± 9.36	49.75 ± 11.21	.4800
1 yr ASES	89.95 ± 13.50	71.43 ± 23.38	72.48 ± 25.26	.0102
1 yr SANE	81.79 ± 23.34	73.00 ± 23.18	66.36 ± 28.84	.1368
1 yr PCS	51.19 ± 5.77	47.21 ± 11.06	45.61 ± 8.81	.1436
1 yr MCS	53.16 ± 7.26	50.08 ± 9.92	49.20 ± 10.33	.3682
Met MCID (%)	19 (100.00)	28 (75.68)	23 (92.00)	.0762
Met SCB (%)	17 (89.47)	22 (59.46)	16 (64.00)	.0658
Met PASS (%)	17 (89.47)	17 (45.95)	12 (48.00)	.0044

ASES, American Shoulder and Elbow Surgeons; MCID, minimum clinically important difference; MCS, Mental Component Score; PASS, patient acceptable symptom state; PCS, Physical Component Score; SANE, Single Assessment Numeric Evaluation; SCB, substantial clinical benefit; TSA, total shoulder arthroplasty.

Bold values indicate statistical significant P value.

Table II – Multivariate logistic regression results for each ASES score threshold.

Parameter	Odds ratio	95% CI	P value
Met ASES MCID			
Male vs. female	1.42	0.53-3.80	.4893
Reverse vs. anatomical	0.82	0.31-2.18	.6862
Medicare advantage vs. Commercial	0.60	0.09-4.05	.5963
Traditional Medicare vs. commercial	0.21	0.05-0.99	.0491
Age	1.03	0.96-1.11	.4460
Met ASES SCB			
Male vs. female	0.85	0.43-1.69	.6461
Reverse vs. anatomical	1.20	0.60-2.40	.6007
Medicare advantage vs. commercial	0.46	0.14-1.56	.2148
Traditional Medicare vs. commercial	0.24	0.09-0.68	.0070
Age	1.04	0.99-1.10	.1374
Met ASES PASS			
Male vs. female	2.38	1.22-4.67	.0114
Reverse vs. anatomical	0.93	0.47-1.82	.8256
Medicare advantage vs. commercial	0.27	0.09-0.83	.0226
Traditional Medicare vs. commercial	0.33	0.13-0.88	.0272
Age	1.04	0.99-1.10	.1072

ASES, American Shoulder and Elbow Surgeons; CI, confidence interval; MCID, minimum clinically important difference; PASS, patient acceptable symptom state; SCB, substantial clinical benefit. Bold values indicate statistical significant P value.

variables such as age, gender, and procedure (Table II). A similar result was observed in SCB achievement, where Traditional Medicare patients were 76% less likely to achieve the threshold ($P = .0070$), and no other findings were significant. The achievement of PASS was more than double in male patients than female patients ($P = .0114$). Both Medicare Advantage and Traditional Medicare were statistically significant associated with decreased likelihood of achieving PASS (73% and 67%; $P = .0226$ and 0.0272 , respectively). No other findings were statistically significant.

Discussion

Although no significant difference was observed in aTSA PROM achievement threshold, significant difference in rTSA achievement of PASS score was observed, with commercial patients associated with a higher rate of achieving PASS. When controlling for potential confounders such as age, gender, and procedure type, Traditional Medicare was associated with lower likelihood of meeting MCID, SCB, and PASS to commercial payer. Thus, adjusted analysis found payer type to be associated with achievement of certain PROM metrics, mainly decreases in likelihood among traditional Medicare patients.

A recent study in both aTSA and rTSA with respect to private (commercial) and Medicare payers found no difference in

preoperative PROMs such as the ASES and simple shoulder test or MCID achievement rates; however, those Commercial payers achieved a statistically significant ~10-point increase on the simple shoulder test.¹⁴ Although the study did not differentiate between procedures (aTSA and rTSA) or Medicare types (traditional and advantage), their finding of no significant difference in achievement rates compare with our unadjusted analysis on percentage of achievers by payer type (barring our result of PASS achievement being significantly higher for commercial rTSA patients). Similar results have been found when differentiating between Health Maintenance Organization (HMO) and Preferred Provider Organization (PPO) insurance plans and in comparison, to government payers such as Medicare and Veteran's Health Administration Healthcare, where all cohorts achieved a mean significant improvement and largely similar postoperative PROMs.¹⁵ However, to our knowledge, the assessment of Medicare subgroup has not been previously reported with respect to PROM threshold achievement.

The authors are of the opinion that payer type operates as a proxy for SES, and although out of the scope of this manuscript, may represent both clinical and nonclinical variables previously associated with postoperative recovery. Orthopedic research on PROMs has repeatedly demonstrated the susceptibility of scores to variables such as Charleston Comorbidity Index, tranexamic acid use, hospital length of stay, race, other orthopedic conditions excluding the study joint, and number of orthopedic visits,^{1,10,18} age, body mass index, income, smoking status, and payer mix.^{2,3} Importantly, SES, which can be approximated with payer type, has also been demonstrated to impact surgical outcomes.^{2,12,20} The sensitivity of MCID and PROMs largely to a multitude of variables may demonstrate the limitations of using such a threshold.

The results and presentation of this study are timely and notable when considering recent measures and legislation in other orthopedic subspecialties indicating the use of PROMs as a determinant of reimbursement by CMS.¹¹ An important aspect of CMS' PROM mandate is using PROMs as quality measurements of both the surgical care and patient selection in efforts to reduce waste and enhance treatment. However, the results of this study illuminate the complexities of assessing diverse and often disparate populations of patients by PROMs and derived thresholds. The SCB was selected in the CMS PROM mandate in joint arthroplasty to serve as a marker of success or failure.¹¹ This study found that, with respect to the SCB achievement rate, Traditional Medicare patients were 76% less likely to achieve the threshold than Commercial payers. Such a finding was not observed in the unadjusted rates unless other variables were controlled. Further, although 1-year postoperative evaluation is the standard collection interval for PROMs, previous studies have demonstrated recovery timelines for rTSA exceeding one year.^{7,17} If a procedure and recovery are assessed at a uniform 1-year follow-up for rTSA, full recovery may not yet be achieved, thus, potentially leading to an underestimation of the true outcome. In the era of value-based care, the risk of inequities in postoperative recovery should be researched further and proactively addressed to provide quality care to all patients. If reimbursement and quality metrics are linked to PROMs and

achievement thresholds, surgeons who treat higher-risk patients could be inadvertently penalized, possibly limiting access to care for patients who are at the most need. Despite this, PROMs remain exceptionally important to the feedback loop that patient perception provides to clinicians, and the standardized administration and collection efforts are also crucial for comparative research and analytic purposes, with national registries aware of and advocating for increased collection rates.³

Limitations

This study is not without limitations. This study was conducted at a single, high-volume urban site, which may affect generalizability to other practices and locations as well as introduce bias. rTSA patients may not be expected to make a full recovery by 1 year. This is an important factor that may contributor to outcomes disparities and should be carefully addressed by policymakers regarding dependence on PROM assessment at set intervals, notably 1 year. As previously mentioned, payer type likely serves as a proxy for SES; however, the explicit interest in focusing on payer type in the analysis is the easy availability of this information by providers. This study demonstrates negative associations with PROM achievement and payer type, which could result in providers being disincentivized from taking on patients from certain payer groups, further exacerbating existing barriers to access of care demonstrated in patients with lower SES. Lastly, we acknowledge that multivariate logistic modeling is limited by low sample sizes in some categories and possibly interdependent variables.

Conclusion

The achievement of clinically relevant PROM thresholds such as the MCID, SCB, and PASS was dissimilar among payer types, with Traditional Medicare patients experiencing significant reductions in likelihood of achieving these important recovery milestones when controlling for potentially relevant factors. Recognizing these trends can help improve and design reimbursement models that promote equitable outcomes and ensure quality metrics reflect care.

Disclaimers:

Funding: The Fondren Orthopedic Research Institute at the Texas Orthopedic Hospital financially supported a portion of the study team.

Conflicts of interest: T.B.E. receives royalties from DJ Orthopedics and Responsive Arthroscopy and is a paid consultant for DJ Orthopedics, Kinamed, Smith & Nephew, United Orthopedic Company, and Wright Medical Technology also receiving research support from Tornier and Wright Medical Technology; however, this did not impact the study. M.C.C. is a paid consultant for Enovis, Medacta and Aevumed, and owns stock in Aevumed; however, this did not impact the study. The other authors, their immediate

families, and any research foundation with which they are affiliated have not received any financial payments or other benefits from any commercial entity related to the subject of this article.

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